Lecture 1

Taylor's polynomial of degree n of f at x_0

$$p_n(x) = f(x_0) + \frac{(x-x_0)}{1!}f'(x_0) + \dots + \frac{(x-x_0)^n}{n!}f^{(n)}(x_0) \quad (1.2)$$

error (1.3)

$$R_{n+1}(x) = \frac{1}{n!}\int^{x}(x-t)^n f^{(n+1)}(t)dt = \frac{(x-x_0)^{n+1}}{(n+1)!}f^{(n+1)}(\xi),\ \xi \text{ between } x \text{ and } x_0$$

Polynomial Interpolation — Lecture 2

Lagrange Interpolation

$$L_n f(x) = \sum_{i=0}^{n} l_i(x) f(x_i) \quad (1.3)$$

$$l_i(x) = \prod_{\substack{j=0\\ j\neq i}}^{n} \frac{x-x_j}{x_i-x_j} = \frac{u_i(x)}{u_i(x_i)} = \frac{u_i(x)}{u'(x_i)} \quad (1.4)$$

exists $\xi \in (a,b)$ $\quad (R_n f)(x) = \frac{u(x)}{(n+1)!}f^{(n+1)}(\xi).$ (1.6)

Nr.	Derivate
1	$c' = 0$
2	$x' = 1$
3	$(x^n)' = nx^{n-1}$
4	$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$
5	$\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$
6	$(e^x)' = e^x$
7	$(a^x)' = a^x \ln a$
8	$(\ln x)' = \frac{1}{x}$
9	$(\log_a x)' = \frac{1}{x \ln a}$
10	$(\text{arctg}\, x)' = \frac{1}{x^2+1}$
11	$(\text{arcctg}\, x)' = -\frac{1}{x^2+1}$
12	$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$
13	$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$
14	$(\sin x)' = \cos x$
15	$(\cos x)' = -\sin x$

Nr.	Integrale nedefinite
1	$\int dx = x + C$
2	$\int x dx = \frac{x^2}{2} + C$
3	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$
4	$\int \sqrt{x} dx = \frac{2}{3}x\sqrt{x} + C$
5	$\int e^x dx = e^x + C$
6	$\int a^x dx = \frac{a^x}{\ln a} + C$
7	$\int \frac{1}{x} dx = \ln\|x\| + C$
8	$\int \frac{1}{x^2-a^2} dx = \frac{1}{2a}\ln\left\|\frac{x-a}{x+a}\right\| + C$
9	$\int \frac{1}{x^2+1} dx = \text{arctg}\, x + C$

Derivate

$(x^n)' = nx^{n-1}$

$(\cos x)' = -\sin x$

$(\log_a x)' = \frac{1}{x \ln a}$

$(a^x)' = a^x \ln a$

$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

$(fg)' = f'g + fg'$

Integrale nedefinite

$\int \ln x dx = x\ln x - x + C$

$\int a^x dx = \frac{a^x}{\ln a} + C$

$\int \sin x\, dx = -\cos x + C$

$\int x^n dx = \frac{x^{n+1}}{n+1} + C$

$\int \frac{dx}{x} = \ln|x| + C$

$\int f(x)f'(x)\,dx = \frac{f^2(x)}{2} + C$

$\int f(x)F(x)\,dx = \frac{F^2(x)}{2} + C$

$\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$

$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + C$

$\int \frac{f(x)}{F(x)}dx = \ln|F(x)| + C$

Birkhoff Interpolation — Lecture 5

$$P^{(j)}(x_k) = f^{(j)}(x_k),\ k = \overline{0,m},\ j \in I_k.$$

$$B_n f(x) = \sum_{k=0}^{m}\sum_{j\in I_k} b_{kj}(x) f^{(j)}(x_k) \quad (1.2)$$

$$\delta_{jp} = \begin{cases} 0, & j \neq p \\ 1, & j = p \end{cases}$$

$$\begin{aligned} b_{kj}^{(p)}(x_\nu) &= 0,\ \nu \neq k,\ p \in I_\nu, \\ b_{kj}^{(p)}(x_k) &= \delta_{jp},\ p \in I_k, \text{ for } j \in I_k \text{ and } \nu, k = 0,1,\dots,m, \end{aligned} \quad (1.3)$$

[Peano] *Peano kernel.*

$$Lf = \int_a^b K_n(t) f^{(n)}(t)dt, \qquad K_n(t) = \frac{1}{(n-1)!}L\left[(\cdot - t)_+^{n-1}\right]$$

$$Lf = \frac{1}{n!}f^{(n)}(\xi)Le_n, \quad (1.13)$$

Divided Differences — Lecture 2

first-order divided difference $\quad f[x_0,x_1] = \frac{f(x_1)-f(x_0)}{x_1-x_0}$ (4.1)

divided difference of order n ***of*** f ***at the distinct nodes*** $x_0,x_1,\dots,x_n$

$$f[x_0,x_1,\dots,x_n] = \frac{f[x_1,x_2,\dots,x_n] - f[x_0,x_1,\dots,x_{n-1}]}{x_n-x_0} \quad (4.3)$$

$$\begin{array}{llllll}
x_0 & f[x_0] & \to f[x_0,x_1] & \to f[x_0,x_1,x_2] & \to f[x_0,x_1,x_2,x_3] \\
x_1 & f[x_1] & \to f[x_1,x_2] & \to f[x_1,x_2,x_3] & \\
x_2 & f[x_2] & \to f[x_2,x_3] & & \\
x_3 & f[x_3] & & &
\end{array}$$

$$f[x_0,x_0,\dots,x_0] = \frac{f^{(n)}(x_0)}{n!} \quad (4.4)$$

divided difference of order n at the node x_0, of multiplicity $n+1$

$$f[x_0,x_1,\dots,x_n] = \sum_{i=0}^{n}\frac{f(x_i)}{u'(x_i)} = \sum_{i=0}^{n}\frac{f(x_i)}{u_i(x_i)}, \quad (4.5)$$

$$u(x) = (x-x_0)(x-x_1)\dots(x-x_n) \text{ and } u_i(x) = \frac{u(x)}{x-x_i}.$$

Finite Differences

$x_i = x_0 + ih,\ i = 0,1,\dots,n,\ h > 0.$

kth-order forward difference of f with step h, at x_i. (4.12)

$$\Delta^k f(x_i) = \Delta^{k-1}f(x_{i+1}) - \Delta^{k-1}f(x_i) = \Delta^{k-1}f_{i+1} - \Delta^{k-1}f_i$$

$$\begin{array}{llllll}
x_0 & f_0 & \to \Delta f_0 & \to \Delta^2 f_0 & \to \Delta^3 f_0 \\
x_1 & f_1 & \to \Delta f_1 & \to \Delta^2 f_1 & \\
x_2 & f_2 & \to \Delta f_2 & & \\
x_3 & f_3 & & &
\end{array}$$

$$\Delta^n f(x_0) = \sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}f_k \quad (4.14)$$

backward difference ∇ (4.13)

$$\nabla^k f_i = \nabla^{k-1}f_i - \nabla^{k-1}f_{i-1},$$

Aitken-type methods — Lecture 3

Neville's method.

$$P_{i,0} := f(x_i),\quad i = \overline{0,n},$$

$$P_{i,j} := \frac{(x-x_{i-j})P_{i,j-1} - (x-x_i)P_{i-1,j-1}}{x_i - x_{i-j}} = \frac{1}{x_i-x_{i-j}}\begin{vmatrix} x-x_{i-j} & P_{i-1,j-1} \\ x-x_i & P_{i,j-1}\end{vmatrix},\quad i \geq j > 0. \quad (1.22)$$

$$\begin{array}{lllll}
x_0 & P_0 & & & \\
x_1 & P_1 & P_{01} & & \\
x_2 & P_2 & P_{12} & P_{012} & \\
x_3 & P_3 & P_{23} & P_{123} & P_{0123}
\end{array}$$

$$P_{i,j} = \tilde{P}_{i-j,i-j+1,\dots,i-1,i},\ j = i, i-1,\dots 0,$$

Aitken's method (1.23)

$$P_{i,0} := f(x_i),\quad i = \overline{0,n},$$

$$P_{i,j+1} := \frac{1}{x_i-x_j}\begin{vmatrix} x-x_j & P_{j,j} \\ x-x_i & P_{i,j}\end{vmatrix} = \frac{(x-x_j)P_{i,j} - (x-x_i)P_{j,j}}{x_i-x_j},\quad i > j \geq 0.$$

Hermite Interpolation — Lecture 4

double nodes $$H_n f(x) = \sum_{i=0}^{m}\left[h_{i0}(x)f(x_i) + h_{i1}(x)f'(x_i)\right] \quad (1.3)$$

$$\begin{aligned} h_{i0}(x) &= \left[1 - 2l_i'(x_i)(x-x_i)\right]\left[l_i(x)\right]^2, \quad (1.4) \\ h_{i1}(x) &= (x-x_i)\left[l_i(x)\right]^2,\ i = 0,\dots,m. \end{aligned}$$

Newton's divided difference form

$$\begin{aligned} N_n(x) &= f(x_0) + f[x_0,x_0](x-x_0) + f[x_0,x_0,x_1](x-x_0)^2 \\ &+ f[x_0,x_0,x_1,x_1](x-x_0)^2(x-x_1) + \dots \quad (1.6) \\ &+ f[x_0,x_0,\dots,x_m,x_m](x-x_0)^2\dots(x-x_{m-1})^2(x-x_m) \end{aligned}$$

$$R_n(x) = f(x) - H_n f(x) = \left[\psi_m(x)\right]^2\frac{f^{(n+1)}(\xi_x)}{(n+1)!},\ \xi_x \in (a,b). \quad (1.9)$$

$$h_{ij}(x) = \frac{(x-x_i)^j}{j!}\left[\sum_{k=0}^{r_i-j}\frac{(x-x_i)^k}{k!}\left[\frac{1}{u_i(x)}\right]_{x=x_i}^{(k)}\right]u_i(x). \quad (1.13)$$

$$\begin{aligned} N_n f(x) &= f(z_0) + f[z_0,z_1](x-z_0) + \dots \quad (1.14) \\ &+ f[z_0,\dots,z_n](x-z_0)\dots(x-z_{n-1}), \end{aligned}$$

$R_n(x)$ (1.15)

$$= \frac{u(x)}{(n+1)!}f^{(n+1)}(\xi_x),\ \xi_x \in (a,b).$$

Least Squares Approximation

$$\begin{aligned} a\sum_{i=i}^{n}x_i^2 + b\sum_{i=i}^{n}x_i &= \sum_{i=i}^{n}x_i y_i \\ a\sum_{i=i}^{n}x_i + nb &= \sum_{i=i}^{n}y_i. \end{aligned}$$

$$\begin{aligned} a\sum_{i=1}^{n}x_i^4 + b\sum_{i=1}^{n}x_i^3 + c\sum_{i=1}^{n}x_i^2 &= \sum_{i=1}^{n}x_i^2 y_i, \\ a\sum_{i=1}^{n}x_i^3 + b\sum_{i=1}^{n}x_i^2 + c\sum_{i=1}^{n}x_i &= \sum_{i=1}^{n}x_i y_i, \\ a\sum_{i=1}^{n}x_i^2 + b\sum_{i=1}^{n}x_i + nc &= \sum_{i=1}^{n}y_i, \end{aligned}$$

Theorem 3.3. *Assume α is a fixed point of g and that g is p times continuously differentiable for all x near α, for some $p \geq 2$. Furthermore, assume that*

$$g'(\alpha) = \dots = g^{(p-1)}(\alpha) = 0,\ g^{(p)}(\alpha) \neq 0. \quad (3.4)$$

Then, if the initial value x_0 is chosen sufficiently close to α, the iteration $x_{n+1} = g(x_n)$ converges to α with order of convergence p, and

$$\lim_{n\to\infty}\frac{x_{n+1}-\alpha}{(x_n-\alpha)^p} = \frac{1}{p!}g^{(p)}(\alpha). \quad (3.5)$$

Theorem 3.3. **[Banach]** *Let $g \in C[a,b]$, with $g([a,b]) \subseteq [a,b]$. Assume that there exists $0 < \lambda < 1$ such that*

$$|g(x) - g(y)| \leq \lambda |x - y|, \ \forall x, y \in [a,b] \ \text{(i.e., } g \text{ is a } \textbf{contraction}\text{)}. \tag{3.2}$$

Then g has a unique fixed point $\alpha \in [a,b]$. Furthermore, the iterates

$$x_{n+1} = g(x_n), \ n \geq 0, \tag{3.3}$$

converge to α, for any choice of $x_0 \in [a,b]$ and the following error estimates hold:

$$\begin{aligned} |x_n - \alpha| &\leq \lambda |x_{n-1} - \alpha|, \ n \geq 1, \\ |x_n - \alpha| &\leq \frac{\lambda^n}{1-\lambda} |x_1 - x_0|. \end{aligned} \tag{3.4}$$

Theorem 3.6. *Assume α is a fixed point of g and that g is continuously differentiable in a neighborhood of α, with*

$$|g'(\alpha)| < 1. \tag{3.8}$$

Then the conclusions of Theorem 3.5 still hold, provided that x_0 is chosen sufficiently close to α.

Theorem 3.5. *Let $g \in C^1[a,b]$, such that $g([a,b]) \subseteq [a,b]$ and*

$$\lambda := \max_{x \in [a,b]} |g'(x)| < 1. \tag{3.6}$$

Then:

a) *Function g has a unique fixed point $\alpha \in [a,b]$.*

b) *For any initial choice $x_0 \in [a,b]$, the sequence $x_{n+1} = g(x_n)$ converges to α, as $n \to \infty$.*

c) $|x_n - \alpha| \leq \lambda^n |x_0 - \alpha| \leq \dfrac{\lambda^n}{1-\lambda}|x_1 - x_0|, \ n \geq 1.$

d)

$$\lim_{n \to \infty} \frac{x_{n+1} - \alpha}{x_n - \alpha} = g'(\alpha), \tag{3.7}$$

so, if $g'(\alpha) \neq 0$, the iterative method $x_{n+1} = g(x_n)$ is linearly convergent to the root α with rate of convergence bounded by λ.

Lecture 7

A_i *are called* ***coefficients***,

$$L(f) = \sum_{i=1}^{m} A_i L_i(f) + R(f), \ f \in X. \quad \ker R = \mathbb{P}_d \iff \begin{cases} R(e_k) = 0, \ k = 0, 1, \ldots, d, \\ R(e_{d+1}) \neq 0, \end{cases}$$

Numerical Differentiation

$$f^{(k)}(\alpha) = \sum_{i=0}^{m} A_i f(x_i) + R(f), \quad \int_a^b f(x)dx = \sum_{k=0}^{m} \sum_{j \in I_k} A_{kj} f^{(j)}(x_k) + R(f),$$

numerical differentiation formula. *numerical integration (quadrature) formula.*

forward difference

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \equiv D_h f(x),$$

$$(RD_h f)(x) = f'(x) - D_h f(x) = -\frac{h}{2} f''(\xi), \ \xi \in (x, x+h).$$

backward difference : $f'(x) \approx \dfrac{f(x) - f(x-h)}{h} \equiv \widetilde{D}_h f(x), \ (R\widetilde{D}_h f)(x) = f'(x) - \dfrac{f(x) - f(x-h)}{h} = \dfrac{h}{2} f''(\xi), \ \xi \in (x-h, x).$

Method of Undetermined Coefficients $D_h^{(2)} f(x) = \dfrac{f(x+h) - 2f(x) + f(x-h)}{h^2}. \ (RD_h^{(2)} f)(x) = f''(x) - D_h^{(2)} f(x) \approx -\dfrac{h^2}{12} f^{(4)}(x).$

Numerical Integration

$$\int_a^b f(x)dx = \sum_{k=0}^{m} A_k f(x_k) + R(f),$$

interpolatory quadrature. $d_{\max} = 2m + 1.$

Trapezoidal Rule

$$\int_a^b f(x)dx = \frac{b-a}{2}\big(f(a) + f(b)\big) - \frac{(b-a)^3}{12} f''(\xi), \ \xi \in (a,b),$$

n subintervals $h = \dfrac{b-a}{n}$. **composite (repeated)**

$$\int_a^b f(x)dx = \frac{h}{2}\Big[f(a) + 2\big(f_1 + \cdots + f_{n-1}\big) + f(b)\Big] - \frac{h^2(b-a)}{12} f''(\xi), \ \xi \in (a,b).$$

Rectangle (Midpoint) Rule

$$\int_a^b f(x)dx = (b-a) f\Big(\frac{a+b}{2}\Big) + R(f),$$

$$R(f) = \frac{f''(\xi)}{2!} \int_a^b \Big(x - \frac{a+b}{2}\Big)^2 dx = \frac{(b-a)^3}{24} f''(\xi), \ \xi \in (a,b).$$

composite (repeated)

$$\int_a^b f(x)dx = h \sum_{i=0}^{n-1} f\Big(a + \big(i + \frac{1}{2}\big)h\Big) + \frac{h^2(b-a)}{24} f''(\xi), \ \xi \in (a,b),$$

Simpson's Rule

$$\int_a^b f(x)dx = \frac{b-a}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right] - \frac{(b-a)^5}{2880} f^{(4)}(\xi), \ \xi \in (a,b)$$

$h = \dfrac{b-a}{2m}$, $n = 2m$ **composite (repeated)**

$$\int_a^b f(x)dx = \frac{h}{3}\Big[f(a) + 4\sum_{i=1}^{m} f_{2i-1} + 2\sum_{i=1}^{m-1} f_{2i} + f(b)\Big] - \frac{h^4(b-a)}{180} f^{(4)}(\xi), \ \xi \in (a,b).$$

Adaptive Quadratures — Lecture 8

$$I \approx \frac{2^p I_{2n} - I_n}{2^p - 1} = I_{2n} + \frac{I_{2n} - I_n}{2^p - 1} \stackrel{\text{not}}{=} R_{2n}$$

Richardson's extrapolation formula.

$$I - I_{2n} \approx \frac{I_{2n} - I_n}{2^p - 1}, \ \textbf{error estimate}$$

Iterated Quadratures

Romberg's Method

$$R_{k,j} = \frac{4^{j-1} R_{k,j-1} - R_{k-1,j-1}}{4^{j-1} - 1}, \ k = \overline{2,n}, \ j = \overline{2,k}.$$

$$\begin{matrix} R_{1,1} & & & & \\ R_{2,1} & R_{2,2} & & & \\ R_{3,1} & R_{3,2} & R_{3,3} & & \\ \vdots & \vdots & \vdots & \ddots & \\ R_{n,1} & R_{n,2} & R_{n,3} & \ldots & R_{n,n} \end{matrix}$$

stopping

$$|R_{n-1,n-1} - R_{n,n}| < \varepsilon. \qquad S_{k,1} = R_{k,2}.$$

Gaussian Quadratures

$$\int_a^b w(x) f(x)\, dx = \sum_{k=1}^{m} A_k f(x_k) + R_m(f)$$

maximum degree of precision, $d = 2m - 1.$

$$\mu_j = \int_a^b w(x) x^j \, dx$$

Gaussian Elimination — Lecture 9

$$x_n = \frac{b_n}{u_{nn}}, \quad \textbf{backward substitution:}$$

$$x_i = \frac{1}{u_{ii}}\Big(b_i - \sum_{j=i+1}^{n} u_{ij} x_j\Big), \ i = \overline{n-1, 1}$$

$$x_1 = \frac{b_1}{l_{11}}, \quad \textbf{forward}$$

$$x_i = \frac{1}{l_{ii}}\Big(b_i - \sum_{j=1}^{i-1} l_{ij} x_j\Big), \ i = \overline{2,n}.$$

partial pivoting (maximal pivoting on columns). $|a_{pk}^{(k)}| = \max\limits_{k \leq l \leq n} |a_{lk}^{(k)}|.$

$(R_i) \longleftrightarrow (R_p)$ **scaled pivoting on columns:**

$$\frac{|a_{pi}|}{s_i} = \max_{1 \leq j \leq n} \frac{|a_{ji}|}{s_j}$$

$$s_i = \max_{j = \overline{1,n}} |a_{ij}| \ \text{or} \ s_i = \sum_{j=1}^{n} |a_{ij}|, \ i = \overline{1,n}.$$

total (maximal) pivoting.

$$|a_{pq}| = \max\{|a_{ij}|, i, j = \overline{k,n}\}$$

$$(R_k) \longleftrightarrow (R_p), \ (C_k) \longleftrightarrow (C_q).$$

$$x' = [x_3 \ x_2 \ x_1]^T.$$

LU Factorization *no row interchanges are necessary*

$A = LU$, **Schur complement** $A' - vw^*/a_{11} = L'U'$. $A = \begin{bmatrix} a_{11} & w^* \\ v & A' \end{bmatrix}$

$$\begin{aligned} Ly &= b \\ Ux &= y. \end{aligned} \quad \begin{bmatrix} 1 & 0 \\ v/a_{11} & I_{n-1} \end{bmatrix} \begin{bmatrix} a_{11} & w^* \\ 0 & A' - vw^*/a_{11} \end{bmatrix}$$

LUP Factorization $PA = LU$. $Ly = Pb$

QR Factorization **Lecture 10** $Ux = y$.

$A = QR$, Q orthogonal and R upper triangular $r_{ii} > 0$,

$Rx = Q^T b$.

Cholesky factorization $A = R^T R$, $\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ 0 & & & a_{nn} \end{bmatrix}$

Iterative Methods — Lecture 10

Jacobi and Gauss-Seidel Methods

$$A = D - L - U, \quad x^{(k+1)} = Tx^{(k)} + c$$

Jacobi iteration, $A = M - N.$

$$M = D, \ N = L + U, \text{ so}$$

$$T_J = D^{-1}(L + U), \ c_J = D^{-1} b.$$

$$x^{(k+1)} = D^{-1}(L + U) x^{(k)} + D^{-1} b, \ k \in \mathbb{N},$$

Gauss-Seidel iteration,

$$M = D - L, \ N = U, \text{ so}$$

$$T_{GS} = (D - L)^{-1} U, \ c_{GS} = (D - L)^{-1} b.$$

$$x^{(k+1)} = (D - L)^{-1} U x^{(k)} + (D - L)^{-1} b,$$

Acceleration methods; SOR Method

$$M = \frac{D}{\omega} - L, \ N = \Big(\frac{1-\omega}{\omega} D + U\Big), \text{ so}$$

$$T_\omega = \Big(\frac{D}{\omega} - L\Big)^{-1} \Big(\frac{1-\omega}{\omega} D + U\Big), \ c_\omega = \Big(\frac{D}{\omega} - L\Big)^{-1} b.$$

$$D = \begin{bmatrix} a_{11} & & & 0 \\ & a_{22} & & \\ & & \ddots & \\ 0 & & & a_{nn} \end{bmatrix} \quad -U = \begin{bmatrix} 0 & a_{12} & \ldots & a_{1n} \\ & 0 & \ldots & a_{2n} \\ & & \ddots & \vdots \\ 0 & & & 0 \end{bmatrix}$$

$$-L = \begin{bmatrix} 0 & & & 0 \\ a_{21} & 0 & & \\ \vdots & & \ddots & \\ a_{n1} & a_{n2} & \ldots & 0 \end{bmatrix}$$

Orthogonal polynomials and recurrence coefficients

Name	Notation	Polynomial	Weight fn.	Interval	α_k	β_k
Legendre	l_m	$[(x^2-1)^m]^{(m)}$	1	$[-1,1]$	0	$\beta_0 = 2$, $\beta_k = (4 - k^{-2})^{-1}, k \geq 1$
Chebyshev 1^{st}	T_m	$\cos(m \arccos x)$	$(1-x^2)^{-\frac{1}{2}}$	$[-1,1]$	0	$\beta_0 = \pi$, $\beta_1 = \frac{1}{2}$, $\beta_k = \frac{1}{4}, k \geq 2$
Chebyshev 2^{nd}	Q_m	$\frac{\sin[(m+1)\arccos x]}{\sqrt{1-x^2}}$	$(1-x^2)^{\frac{1}{2}}$	$[-1,1]$	0	$\beta_0 = \frac{\pi}{2}$, $\beta_k = \frac{1}{4}, k \geq 1$
Laguerre	L_m^a	$x^{-a} e^x (x^{m+a} e^{-x})^{(m)}$	$x^a e^{-x}, \ a > -1$	$[0,\infty)$	$2k + a + 1$	$\beta_0 = \Gamma(1+a)$, $\beta_k = k(k+a), k \geq 1$
Hermite	H_m	$(-1)^m e^{x^2} \left(e^{-x^2}\right)^{(m)}$	e^{-x^2}	$\mathbb{R}$	0	$\beta_0 = \sqrt{\pi}$, $\beta_k = \frac{k}{2}, k \geq 1$

$f : [a, b] \to \mathbf{R}$ is continuous c in (a, b)
g is an integrable function
that does not change sign on $[a, b]$.

$$\int_a^b f(x) g(x)\, dx = f(c) \int_a^b g(x)\, dx$$

Common Rootfinding Methods — Lecture 11

Bisection Method

two-point method, $c_n = \dfrac{a_n + b_n}{2}$

Secant Method *two-point method,*

$$x_{n+1} = x_n - f(x_n) \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}, \ n = 1, 2, \ldots,$$

Newton's Method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \ n = 0, 1, \ldots. \quad \textit{one-step}$$

Lecture 12

Newton's Method for Multiple Roots

$$x_{n+1} = x_n - m \frac{f(x_n)}{f'(x_n)}.$$

Newton's Method for Nonlinear Systems

$$\boldsymbol{x_{n+1}} = \boldsymbol{x_n} - \big(\boldsymbol{J_f(x_n)}\big)^{-1} \boldsymbol{f(x_n)}, \ n \geq 0,$$

$$\boldsymbol{J_f(x_n)} = \begin{bmatrix} \dfrac{\partial f_1(\boldsymbol{x})}{\partial x_1} & \dfrac{\partial f_1(\boldsymbol{x})}{\partial x_2} \\ \dfrac{\partial f_2(\boldsymbol{x})}{\partial x_1} & \dfrac{\partial f_2(\boldsymbol{x})}{\partial x_2} \end{bmatrix} \quad \begin{aligned} \boldsymbol{J_f(x_n)\delta_{n+1}} &= -\boldsymbol{f(x_n)}, \\ \boldsymbol{x_{n+1}} &= \boldsymbol{x_n} + \boldsymbol{\delta_{n+1}}. \end{aligned}$$

$$\mathrm{tr}(A) = \sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \lambda_i = \lambda_1 + \lambda_2 + \cdots + \lambda_n$$

$$\det(A) = \prod_{i=1}^{n} \lambda_i = \lambda_1 \lambda_2 \cdots \lambda_n$$

[Banach] (3.2)
$g \in C[a,b]$, *with* $g([a,b]) \subseteq [a,b]$
there exists $0 < \lambda < 1$
$|g(x) - g(y)| \leq \lambda |x - y|, \forall x, y \in [a,b]$
g *has a unique fixed point* $\alpha \in [a,b]$